

AJKD Atlas of Renal Pathology: Immunotactoid Glomerulopathy

Agnes B. Fogo, MD,¹ Mark A. Lusco, MD,¹ Behzad Najafian, MD,² and Charles E. Alpers, MD²

Clinical and Pathologic Features

Immunotactoid glomerulopathy occurs in adults, on average at 60 years of age, and often presents with nephrotic proteinuria, reduced glomerular filtration rate, and hypocomplementemia in half of affected patients. Patients do not have cryoglobulins, but a circulating monoclonal paraprotein and/or lymphoplasmacytic malignancy is present in about two-thirds. Prognosis depends on the outcome of the associated disease, with half of the affected patients achieving remission with therapy directed at the malignancy. This rarely leads to end-stage kidney disease. Recurrence in transplanted kidneys has been reported.

Light microscopy: Variable appearance, most commonly a membranoproliferative pattern or less frequently membranous pattern, and occasionally endocapillary proliferation.

Immunofluorescence microscopy: Mesangial and chunky capillary wall immunoglobulin G, which is often clonal, and less C3, with a light chain restriction. Granular capillary loop staining is seen in cases with a membranous pattern.

Electron microscopy: Microtubules, often in parallel arrays, with diameter typically >30 nm.

Etiology/Pathogenesis

Immunotactoid glomerulopathy is associated with underlying lymphoplasmacytic disorders, and often has clonal deposits (ie, restricted to one light chain subclass) supporting a causal link of the deposits with the underlying malignancy.

Differential Diagnosis

Immunotactoid glomerulopathy is distinguished from fibrillary glomerulonephritis by the larger microtubular substructure revealed by electron microscopy, and the often clonal staining. Cryoglobulinemic glomerulonephritis is most often

immunoglobulin M dominant, can also be clonal, and has deposits with microtubular or short, vague fibrillary substructure revealed by electron microscopy. Type I monoclonal cryoglobulin deposits may show an appearance indistinguishable from immunotactoid glomerulopathy, and the more common mixed cryoglobulinemic deposits also sometimes can show parallel arrays or microtubules. Some other entities with substructure of deposits revealed by electron microscopy do not show immunofluorescence positivity for immunoglobulins, such as fibronectin glomerulopathy or collagenofibrotic glomerulopathy.

Key Diagnostic Features

- Mesangial proliferation and variable endocapillary proliferation
- Often clonal IgG and light chain restriction, and less C3
- Microtubules in parallel arrays in mesangium, and variably in glomerular basement membrane, >30 nm in diameter

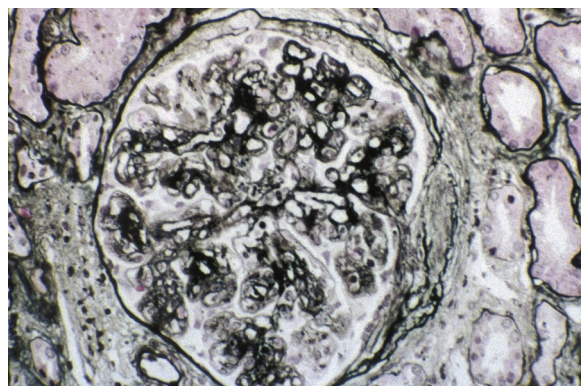


Figure 1. Immunotactoid glomerulopathy showing membranoproliferative pattern with widespread glomerular basement membrane splitting and mesangial matrix expansion, and an adhesion to Bowman capsule (Jones silver stain). Reproduced with permission from *AJKD* 41(1):e23.

From the ¹Department of Pathology, Microbiology and Immunology, Vanderbilt University, Nashville, TN; and ²Department of Pathology, University of Washington, Seattle, WA.

Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

Address correspondence to agnes.fogo@vanderbilt.edu

Am J Kidney Dis. 66(4):e29-e30.

© 2015 by the National Kidney Foundation, Inc.

0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2015.08.003>

Continued

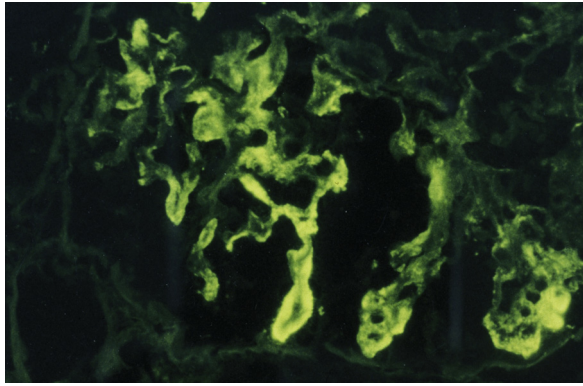


Figure 2. Immunotactoid glomerulopathy showing segmental mesangial and capillary loop positivity, mirroring the distribution of deposits by light microscopy (immunofluorescence microscopy, IgG). Reproduced with permission from *AJKD* 41(1):e23.

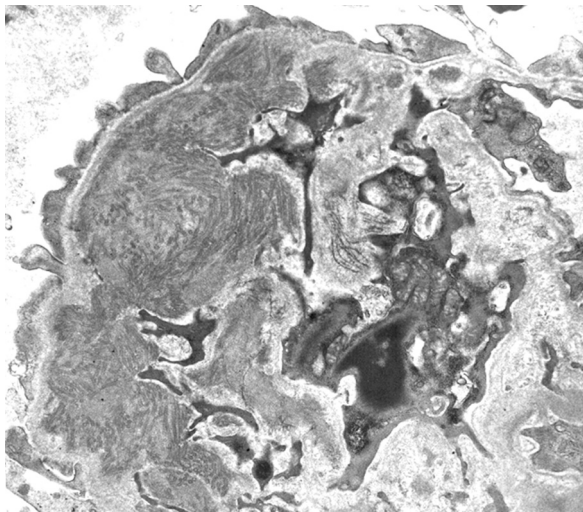


Figure 3. Immunotactoid glomerulopathy with organized deposits in parallel arrays and with microtubular substructure seen in cross-section, approximately 35 nm in diameter (electron microscopy).

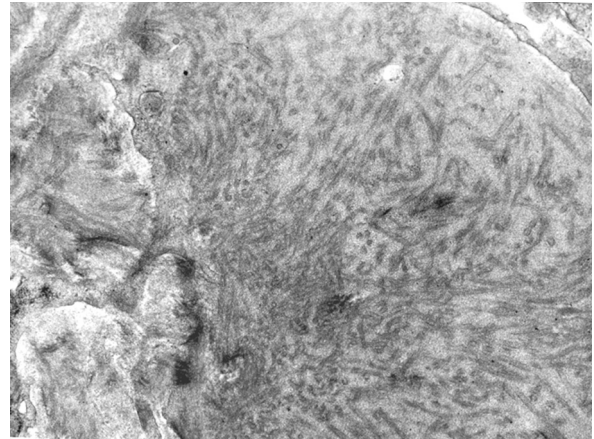


Figure 4. Immunotactoid glomerulopathy again with organized deposits showing stacking in parallel arrays and with microtubular substructure seen in cross-section cut (electron microscopy).